

ECG Image Digitization

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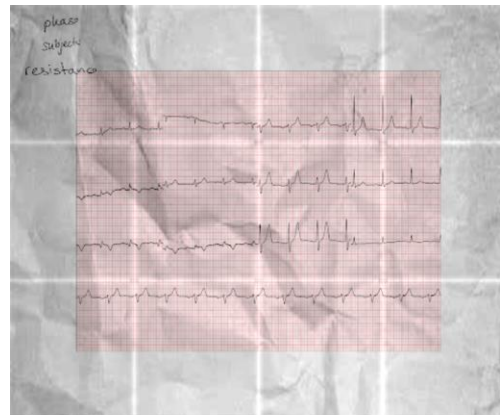
Background

- ▶ **Electrocardiogram (ECG)**
 - Records electrical activity of the heart
 - Helps with diagnoses of heart rhythm diseases and heart attacks
- ▶ **ECGs are often printed on ECG thermal paper**
 - Reliable, Standardized, Widely Used
 - Still used for a variety of reasons:
 - Familiarity
 - Convenience
 - Ease of Interpretation
- ▶ **Increased need for digitalization**
 - Digital/Electronic health records & storage
 - Potential support for AI/ML analysis



Data Sources and Format

- ▶ There are 2 data sources we will need:
 - ▶ **Ground Truth Labels:** Digitized ECG waveforms from real patients
 - ▶ PTB-XL has 21,799 12-lead ECG recordings (~18,885 patients) we can use
 - ▶ **Training Input:** Images rendered from digital waveforms (with realistic creases, rotation, noise)
 - ▶ ECG-Image-Kit is a tool for converting digital ECG to realistically degraded images
- ▶ We can choose to create our own training inputs using ECG-Image-Kit or use existing versions:
 - ▶ PTB-XL-Image-17K has 17K synthetic images
 - ▶ ECG-Image-Database has some real degraded scans (soaked/stained/photographed)
 - ▶ PTB-Image has 549 real scanned images (smaller scale for initial testing)



(a) Wrinkles and creases



(b) Crease artifacts

Target Output

- ▶ **Target:** Use the images to reconstruct a true 12 lead ECG signal
 - ▶ WFDB format
 - ▶ Judged on signal fidelity metrics (SNR, QRS, etc.)
 - ▶ As close as possible to the ground truth signal
- ▶ From existing sources, the pipeline is as follows:
 - ▶ Grid detection to establish pixel-to-(time, voltage) scale
 - ▶ Segmentation to isolate signal from grid/noise (U-Net handles overlapping leads)
 - ▶ Signal extraction to create a 1D waveform mask
 - ▶ Check against expected ECG morphology
- ▶ Existing methods were used in the 2024 PhysioNet Challenge

Previous Attempts

- ▶ Traditional image processing to remove ECG grids & background noise
- ▶ MATLAB tools to convert scanned paper ECGs into digital signals^[1]
 - ~95% precision
- ▶ Automated methods exist to separate the 12 leads & reconstruct individual wave forms^[2]
- ▶ Newer methods use segmentation, perception correction to handle photos in addition to scans^[3]
- ▶ **Challenges still exist in overlaps, perspective, creases, blur, shadows, & other image imperfections^[3,4]**

Next Steps for Investigation

- ▶ Replicate existing digitization techniques for images of paper ECG files

Fig. 8: The digitization pipeline consists of semantic segmentation, that separates gridlines, ecg signal, text and background, followed by perspective correction and cropping.

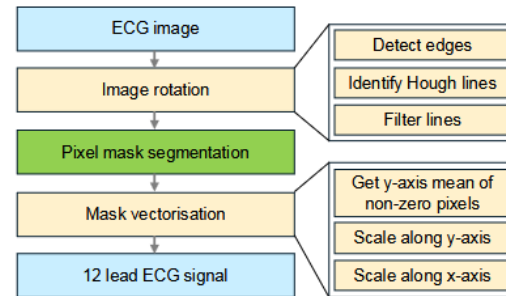
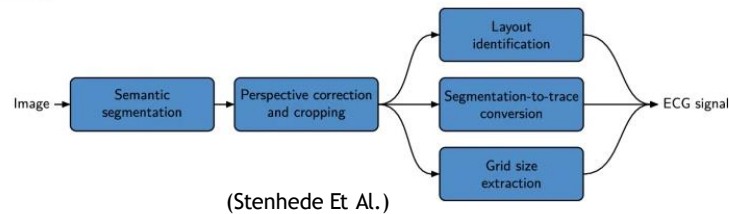


Figure 2. A schematic diagram of our model architecture. Blue: Data, Yellow: Rule-based engineering, Green: Deep learning.

(Krones Et AL.)

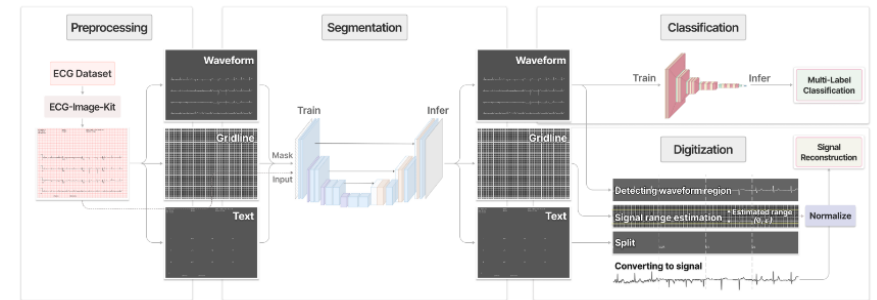


Figure 1. The methodology for extracting key components from generated ECG images using a segmentation model, including the requisite preprocessing steps. These components are employed in the digitization and classification processes.

(Yoon Et AL.)

- ▶ The details and order of individual steps differ, but all approaches do the following:
 - ▶ Extracting the waveform, gridline, and text
 - ▶ Transforming the image to eliminate distortions
 - ▶ Digitizing the waveform
 - ▶ Training the model to maximize signal/noise ratio of the digitized waveform
- ▶ We plan to attempt to replicate these methods with our data before exploring how we can improve them

Compute Requirements

- ▶ **Dataset & Storage**
 - ▶ **Small training set:** 214 GiB
 - ▶ **Large training set:** 1.2 TiB Raw physiological signals are stored in **EDF format**, including ECG
- ▶ **CPU — Initial Processing & Baseline**
 - ▶ ECG data extraction and preprocessing
 - ▶ Classical image processing
 - ▶ Thresholding and line detection
 - ▶ Waveform tracing and reconstruction
- ▶ **GPU — Deep Learning Experiments**
 - ▶ ECG trace segmentation / signal-recovery models
 - ▶ Image augmentation and model training
- ▶ **Google Colab** for initial GPU experiments
- ▶ **JHU GPU cluster** for larger-scale experiments, if needed

Thank you for listening

Any Questions?

References

- [1]: Baydoun, M., Safatly, L., Abou Hassan, O. K., Ghaziri, H., El Hajj, A., & Isma'eel, H. (2019). High Precision Digitization of Paper-Based ECG Records: A Step Toward Machine Learning. IEEE journal of translational engineering in health and medicine, 7, 1900808. <https://doi.org/10.1109/JTEHM.2019.2949784>
- [2]: Wu, H., Patel, K.H.K., Li, X. et al. A fully-automated paper ECG digitisation algorithm using deep learning. Sci Rep 12, 20963 (2022). <https://doi.org/10.1038/s41598-022-25284-1>
- [3]: Stenhede, E., Bjørnstad, A. M., & Ranjbar, A. (2026). Digitizing paper ECGs at scale: an open-source algorithm for clinical research. NPJ digital medicine, 9(1), 145. <https://doi.org/10.1038/s41746-025-02327-1>
- [4]: George S. Waits, Elsayed Z. Soliman, Digitizing paper electrocardiograms: Status and challenges, Journal of Electrocardiology, Volume 50, Issue 1, 2017, Pages 123-130, ISSN 0022-0736, <https://doi.org/10.1016/j.jelectrocard.2016.09.007>.
- [5]: H. Krones, F., J. Lyons, T., Mahdi, A. & Walker, B. Combining hough transform and deep learning approaches to reconstruct ECG signals from printouts. In Computing in Cardiology, <https://www.cinc.org/archives/2024/pdf/CinC2024-097.pdf> (2024).
- [6]: Yoon, H.-C. et al. Segmentation-based extraction of key components from ECG images: a framework for precise classification and digitization. In Computing in Cardiology, <http://www.cinc.org/archives/2024/pdf/CinC2024-227.pdf> (2024).